Exercise 14:

⇒ Parameterizing Earth Ellipsoid

- An ellipsoid centered at $(0,0,0)$ with semi-axes $a=b=R_{eq}$ & $c=R_{pole}$ is defined by:

$$\frac{x^2}{R_{eq}^2}+\frac{y^2}{R_{eq}^2}+\frac{z^2}{R_{pole}^2}=1$$

where

R_{eq} = Earth's equatorial radius = 6378137[m]

$R_{pole}=R_{eq}(1-f)$ = Earth's polar radius

f = flattening = 1/298.2572

- Can parameterize ellipsoid via...

1. Spherical angular parameterization

$$x=R_{eq}\sin(\beta)\cos(\theta)$$
$$y=R_{eq}\sin(\beta)\sin(\theta)$$
$$z=R_{pole}\sin(\beta)$$

where

Azimuth angle θ: $[0,2\pi)$

Elevation angle β: $[-\pi/2,\pi/2]$

2. Geodetic Parameterization

$$x=N(\phi)\cos(\phi)\cos(\lambda)$$
$$y=N(\phi)\cos(\phi)\sin(\lambda)$$
$$z=N(\phi)(1-e^2)\sin(\phi)$$

where

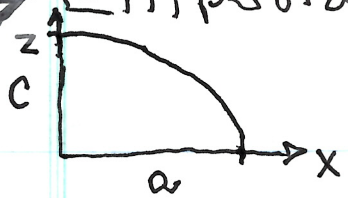
$e=\sqrt{1-(1-f)^2}$ = first eccentricity

$N(\phi)=\dfrac{R_{eq}}{\sqrt{1-e^2\sin^2(\phi)}}$ = radius of curvature in prime vertical

ϕ = geodetic latitude

λ = longitude

⇒ Ellipsoid Geometry Definitions;



a, b, c are semi-axes

- Flattening: $f = \frac{a-c}{a} = 1 - \frac{c}{a} \rightarrow \frac{c}{a} = 1 - f$

First eccentricity: $e = \sqrt{1 - c^2/a^2}$

$$= \sqrt{1 - (c/a)^2}$$
$$= \sqrt{1-(1-f)^2}$$
$$e^2 = 1 - (1 - 2f + f^2)$$
$$= 2f - f^2$$

f in terms of e given by:

$$e^2 = 1 - (1-f)^2$$
$$-e^2 + 1 = (1-f)^2$$
$$1 - f = \sqrt{1 - e^2}$$
$$f = 1 - \sqrt{1 - e^2}$$

⇒ Coordinate Systems & Ref. Frames

- Reference frame: specified by an ordered set of three mutually orthogonal, possibly time dependent, unit length direction vectors; has an associated center
- Coordinate system: specifies a mechanism for locating points within a reference frame

⇒ Converting LLA to ECEF:

ϕ = geodetic latitude

λ = longitude

h = height above reference ellipsoid

$R_{eq} = 6378.135$ [km], $R_p = R_{eq}(1-f)$

$f = 1/298.26$

$e^2 = 2f - f^2$

1. Calculate radius of curvature in prime vertical:

$$N(\phi) = \frac{R_{eq}}{\sqrt{1-e^2\sin^2(\phi)}}$$

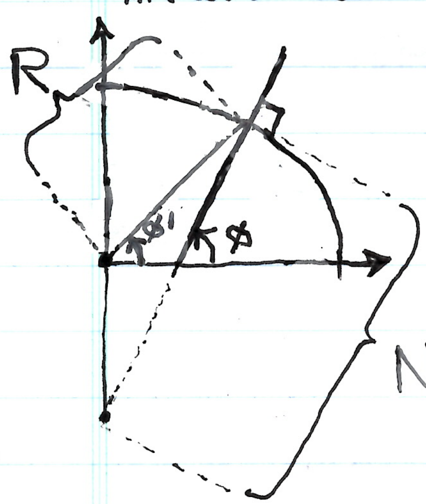
2. Compute ECEF coordinates

$$X = (N(\phi)+h)\cos(\phi)\cos(\lambda)$$

$$Y = (N(\phi)+h)\cos(\phi)\sin(\lambda)$$

$$Z = (N(\phi)(1-e^2)+h)\sin(\phi)$$

* Note: geodetic latitude (ϕ) is measured relative to the normal (perpendicular) of the reference ellipsoid, whereas geocentric latitude (ϕ') is measured from the center of Earth



$R(\phi)$ = geocentric radius

$$= \sqrt{\frac{(R_{eq}^2\cos(\phi))^2 + (R_p^2\sin(\phi))^2}{(R_{eq}\cos(\phi))^2 + (R_p\sin(\phi))^2}}$$

= straight line distance from Earth's center to a point on the surface

$$R(\phi') = \frac{R_{eq}(1-f)}{\sqrt{1-(2f-f^2)\cos^2(\phi')}}$$

ECEF to LLA Conversion Algorithm

Converting Cartesian coordinates (X, Y, Z) in an Earth-Centered, Earth-Fixed (ECEF) frame to geodetic coordinates—Latitude (ϕ), Longitude (λ), and Altitude (h)—requires accounting for the Earth's ellipsoidal shape rather than assuming a perfect sphere.

1. Ellipsoid Constants (WGS 84)

The conversion relies on the reference ellipsoid parameters. For the standard **WGS 84** model:

- **Semi-major axis (a):** 6378137.0 m
- **Flattening (f):** $\frac{1}{298.257223563}$
- **Semi-minor axis (b):** $b = a(1 - f) \approx 6356752.3142 \text{ m}$
- **First eccentricity squared (e^2):** $e^2 = \frac{a^2 - b^2}{a^2} = 2f - f^2 \approx 0.00669437999014$
- **Second eccentricity squared (e'^2):** $e'^2 = \frac{a^2 - b^2}{b^2} = \frac{e^2}{1 - e^2} \approx 0.00673949674227$

2. Closed-Form Algorithm (Bowring's Method)

While exact solutions exist via Ferrari's method, **Bowring's algorithm** (1976/1985) is the industry standard for aerospace and navigation applications. It is non-iterative and accurate to sub-millimeter levels anywhere near or above the Earth's surface.

Step 1: Distance from Z-axis

Calculate the planar distance p from the Earth's rotation axis:

$$p = \quad X^2 + Y^2$$

Step 2: Geodetic Longitude (λ)

Longitude is computed directly:

$$\lambda = \text{atan2}(Y, X)$$

(Note: Use `atan2` to ensure the correct quadrant output between $-\pi$ and π.)

Step 3: Auxiliary Parametric Angle (θ)

Compute the reduced latitude angle θ:

$$\theta = \text{atan2}(a \cdot Z,\ b \cdot p)$$

Step 4: Geodetic Latitude (ϕ)

Using Bowring's formula:

$$\phi = \text{atan2}\left(Z + e'^2 \cdot b \cdot \sin^3\theta,\ \ p - e^2 \cdot a \cdot \cos^3\theta\right)$$

Step 5: Ellipsoidal Altitude (h)

First, find the **prime vertical radius of curvature** $N(\phi)$:

$$N(\phi) = \frac{a}{1 - e^2 \sin^2\phi}$$

Then compute height above the ellipsoid h:

$$h = \frac{p}{\cos\phi} - N(\phi)$$

> **Numerical Tip:** Near the Earth's poles ($\phi \to \pm 90^\circ$), $\cos\phi \to 0$. To prevent numerical instability at extreme latitudes, use:
>
> $$h = \frac{Z}{\sin\phi} - N(\phi)(1 - e^2)$$

⇒ Calc. Julian Date from a UT1 Date-time

This converts a Gregorian calendar date (Year Y, Month M, Day D) & time (Hour H, Minute Min, Second S) into a Julian Date (JD)

1. If month is January (1) or February (2), treat it as months 13 & 14 of prev. year
   If $M \leq 2$:
   $Y = Y - 1$
   $M = M + 12$
2. Calc. Gregorian century corrections
   Determine adjustments needed for Gregorian calendar leap year system
   $A = \lfloor Y/100 \rfloor$
   $B = 2 - A + \lfloor A/4 \rfloor$
3. Compute the fractional day
   $$\text{Day fraction} = \frac{H + \frac{Min}{60} + \frac{S}{3600}}{24}$$
4. Final summation
   $JD = \lfloor 365.25(Y+4716) \rfloor + \lfloor 30.6001(M+1) \rfloor$
   $\ldots + D + \text{Day fraction} + B - 1524.5$

→ Calc. GMST from a Julian Date

1. Calc. # of Julian centuries that have ellapsed since J2000.0

$JD_{J2000} = 2451545.0$

$T = (JD - JD_{J2000}) / 36525$

2. Calc. the sidereal angle in degrees using the IAU polynomial expression

$\theta = 280.46061837 + 360.98564736629 * (JD - JD_{J2000})$

$\ldots + 0.000387933 * T^2 - T^3/38710000.0$

3. Normalize angle to value in range [0, 360)

$\theta = \theta \% 360.0$

if $\theta < 0$:

  θ += 360.0

⇒ Converting ECEF to ECI

- Both pos. & vel. need to be converted since the ECEF frame is fixed to & rotating w/ Earth whereas the ECI frame is fixed relative to the stars, w/ its x-axis pointing toward the vernal equinox
- Position conversion:

$$\vec{r}_{ECI} = \begin{bmatrix} x \\ y \\ z \end{bmatrix}_{ECI} = \begin{bmatrix} \cos(\theta_{GST}) & -\sin(\theta_{GST}) & 0 \\ \sin(\theta_{GST}) & \cos(\theta_{GST}) & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix}_{ECEF}$$

where

θ_{GST} = Greenwich Sidereal Time

JD_{UT1} = continuous count of days & fractions of a day elapsed since noon Universal Time on Jan 1, 4713 BC, measured on UT1 time scale

See 'get_jd' fxn in 'coordinate_conversion_tools.py' for the calc. of JD_{UT1}

- Velocity conversion:

$$V_{ECI} = R_z(-\theta_{GST})\vec{V}_{ECEF} + \vec{\omega}_{\oplus} \times \vec{r}_{ECI}$$

$$= \begin{bmatrix} \cos(\theta_{GST}) & -\sin(\theta_{GST}) & 0 \\ \sin(\theta_{GST}) & \cos(\theta_{GST}) & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} V_x \\ V_y \\ V_z \end{bmatrix}_{ECEF}$$

$$\ldots + \begin{bmatrix} 0 \\ 0 \\ \omega_{\oplus} \end{bmatrix} \times \begin{bmatrix} x \\ y \\ z \end{bmatrix}_{ECI}$$

where $\omega_{\oplus} = 7.2921158559 * 10^{-5}$ [rad/s]

⇒ Converting ECI to ECEF

$$\vec{V}_{ECEF} = R(\theta_{GST})(\vec{V}_{ECI} - \vec{\omega}_{\oplus} \times \vec{r}_{ECI})$$

where

$$R(\theta_{GST}) = \begin{bmatrix} \cos(\theta_{GST}) & \sin(\theta_{GST}) & 0 \\ -\sin(\theta_{GST}) & \cos(\theta_{GST}) & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$\vec{V}_{ECI} - \vec{\omega}_{\oplus} \times \vec{r}_{ECI} = \begin{bmatrix} V_x \\ V_y \\ V_z \end{bmatrix}_{ECI} - \begin{bmatrix} 0 \\ 0 \\ \omega_{\oplus} \end{bmatrix} \times \begin{bmatrix} x \\ y \\ z \end{bmatrix}_{ECI}$$

$$\vec{r}_{ECEF} = R(\theta_{GST})\vec{r}_{ECI}$$

$$= \begin{bmatrix} \cos(\theta_{GST}) & \sin(\theta_{GST}) & 0 \\ -\sin(\theta_{GST}) & \cos(\theta_{GST}) & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix}_{ECI}$$